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Activity-4

1.Amounts

Employee Salary Visulation

R CODE:

```
# EMPLOYEE DATASET
```

```
emp <- data.frame(  
  Department = c("HR","IT","Sales","Finance","Marketing"),  
  Male = c(12,20,18,10,15),  
  Female = c(10,15,12,8,11),  
  Salary = c(35000,60000,45000,55000,40000)  
)
```

```
# 1. BAR CHART
```

```
barplot(emp$Salary,  
  names.arg=emp$Department,  
  col="skyblue",  
  main="Employee Salary by Department",  
  xlab="Department",  
  ylab="Salary")
```

```
# 2. STACKED BAR CHART
```

```
emp_count <- rbind(emp$Male, emp$Female)  
barplot(emp_count,  
  names.arg=emp$Department,  
  col=c("orange","green"),  
  main="Male and Female Employees",  
  xlab="Department",  
  ylab="Number of Employees",
```

```

legend=c("Male","Female"))

# 3. HEAT MAP

emp_matrix <- as.matrix(emp[,2:4])

rownames(emp_matrix) <- emp$Department

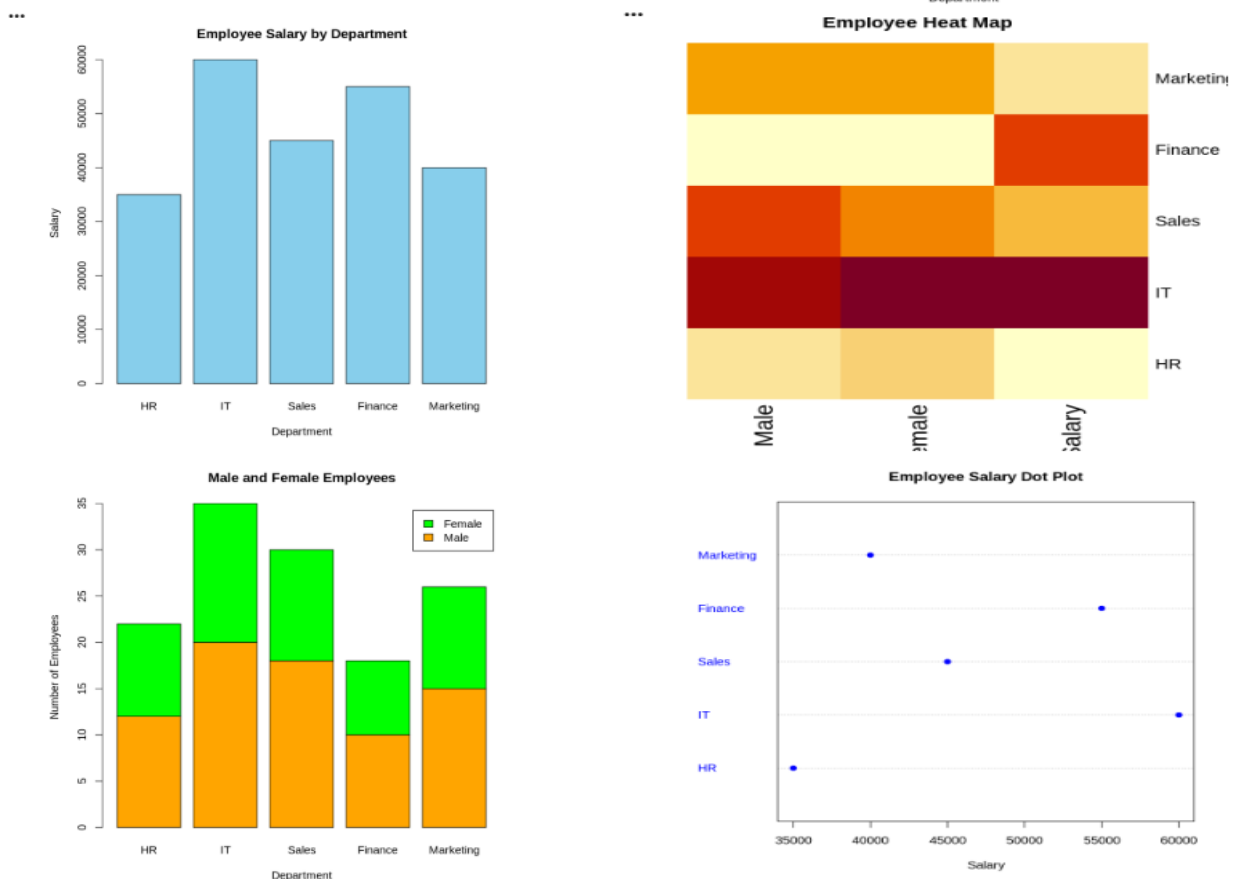
heatmap(emp_matrix,
        Rowv=NA,
        Colv=NA,
        scale="column",
        main="Employee Heat Map")

# 4. DOT PLOT

dotchart(emp$Salary,
        labels=emp$Department,
        main="Employee Salary Dot Plot",
        xlab="Salary",
        pch=19,
        col="blue")

```

Output:



2. Visulation distribution.

```
install.packages("ggplot2")
install.packages("ggridges")
install.packages("vioplot")

# Load libraries
library(ggplot2)
library(ggridges)
library(vioplot)

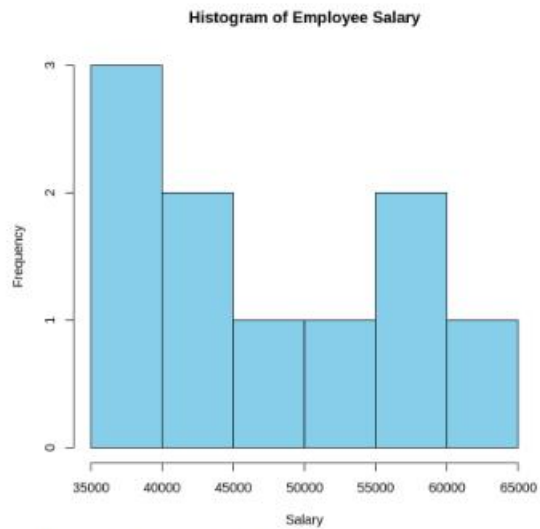
# Employee Dataset
emp <- data.frame(
  Department = c("HR", "IT", "Sales", "Finance", "Marketing",
    "HR", "IT", "Sales", "Finance", "Marketing"),
  Salary = c(35000, 60000, 45000, 55000, 40000,
    36000, 62000, 47000, 57000, 42000)
)

# 1. HISTOGRAM
hist(emp$Salary,
  main="Histogram of Employee Salary",
  xlab="Salary",
  col="skyblue",
  border="black")

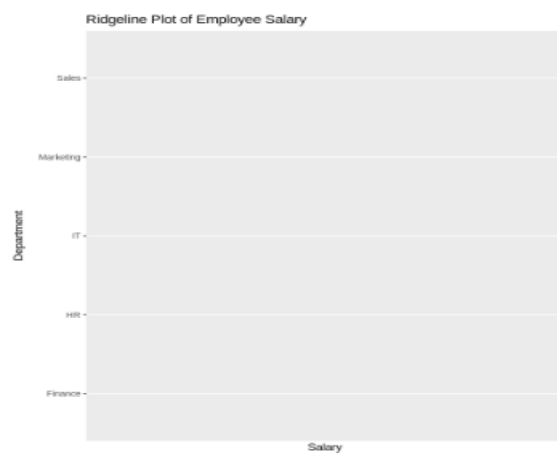
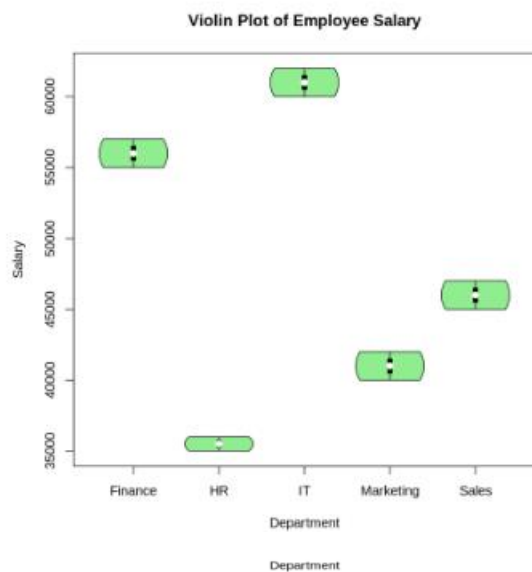
# 2. VIOLIN PLOT
vioplot(Salary ~ Department,
  data=emp,
  col="lightgreen",
  main="Violin Plot of Employee Salary")

# 3. RIDGELINE PLOT
ggplot(emp, aes(x=Salary, y=Department, fill=Department)) +
```

```
geom_density_ridges(alpha=0.7) +
labs(title="Ridgeline Plot of Employee Salary")
```



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3. Visulation properties

Install packages (Run only once)

```
install.packages("ggplot2")
```

```
install.packages("treemap")
```

```
install.packages("hexbin")
```

Load libraries

```
library(ggplot2)
```

```
library(treemap)
```

```
library(hexbin)
```

Employee Dataset

```
emp <- data.frame(  
  Department = c("HR", "IT", "Sales", "Finance", "Marketing"),  
  Employees = c(22, 35, 30, 18, 26),  
  Salary = c(35000, 60000, 45000, 55000, 40000),  
  Experience = c(2, 6, 4, 7, 3)  
)
```

1. BUBBLE CHART

```
ggplot(emp, aes(x=Experience, y=Salary, size=Employees, color=Department)) +  
  geom_point(alpha=0.7) +  
  ggtitle("Bubble Chart")
```

2. TREEMAP

```
treemap(emp,  
  index="Department",  
  vSize="Employees",  
  vColor="Salary",  
  type="value",  
  title="Employee Treemap")
```

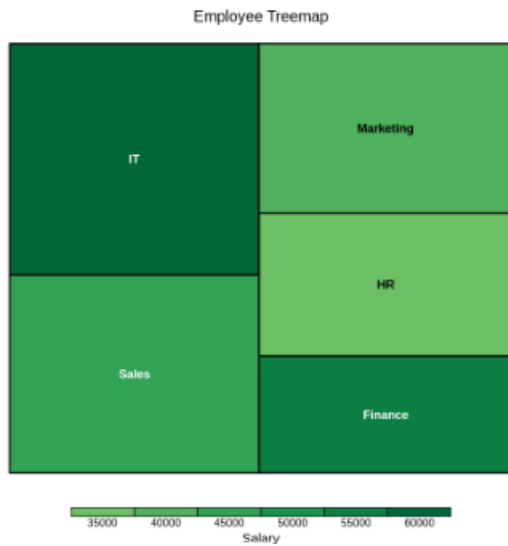
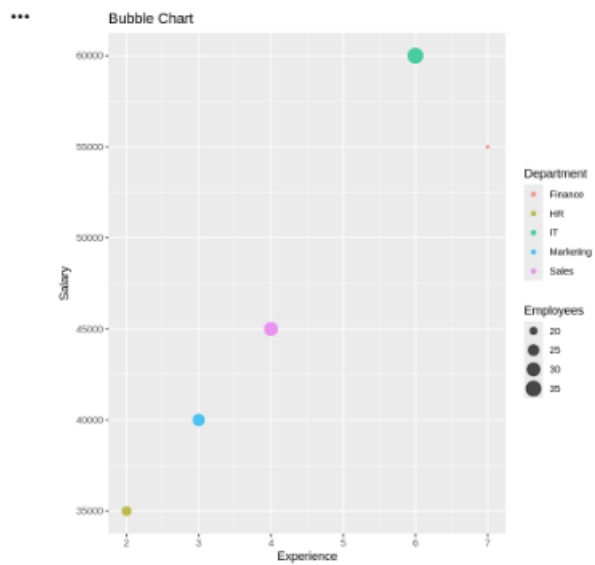
3. X-Y RELATION (SCATTER PLOT)

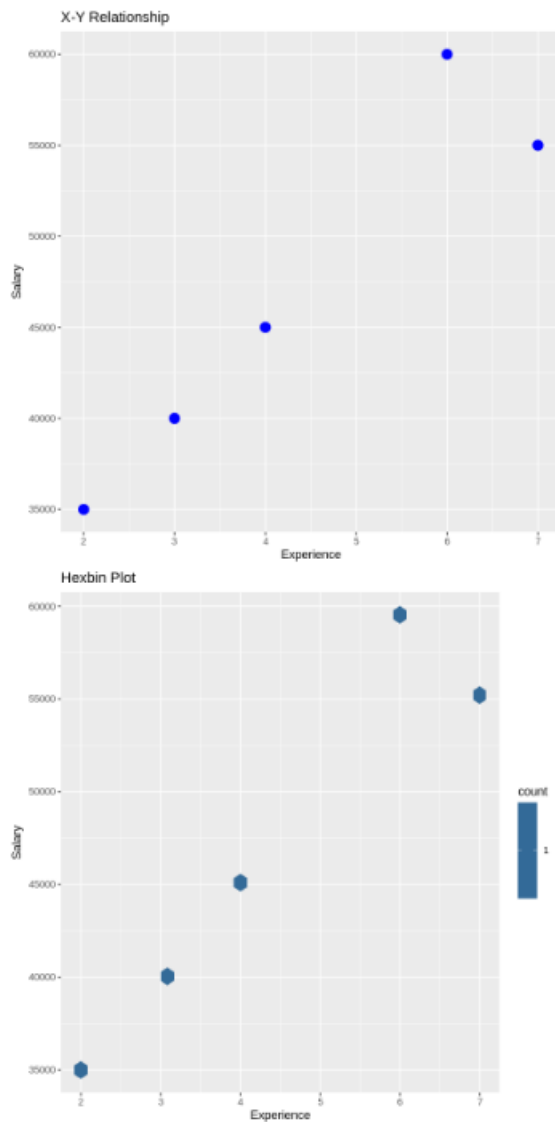
```
ggplot(emp, aes(x=Experience, y=Salary)) +
```

```
geom_point(size=4, color="blue") +
ggtitle("X-Y Relationship")
```

4. HEXBIN PLOT

```
ggplot(emp, aes(x=Experience, y=Salary)) +
  geom_hex() +
  ggtitle("Hexbin Plot")
```





5.Geospatial Data:

Install package (Run only once)

```
install.packages("maps")
```

Load package

```
library(maps)
```

Employee Dataset

```
emp <- data.frame(
```

```
  Experience = c(2,4,6,8,10),
```

```
  Salary = c(30000,40000,50000,60000,70000),
```

```
  Employees = c(20,35,25,40,30)
```

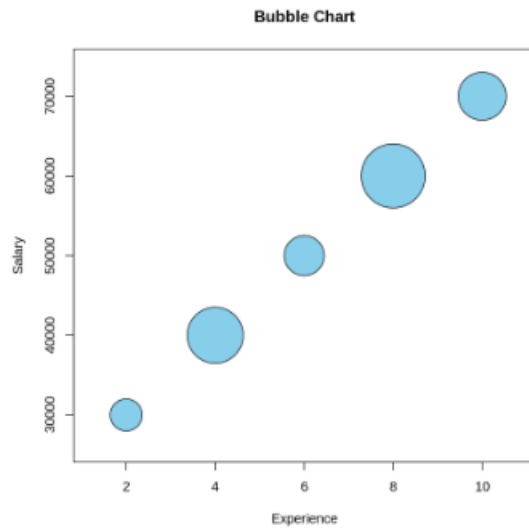
```
)
```

1. BUBBLE CHART

```
symbols(emp$Experience,  
        emp$Salary,  
        circles = emp$Employees/10,  
        inches = 0.4,  
        bg = "skyblue",  
        fg = "black",  
        xlab = "Experience",  
        ylab = "Salary",  
        main = "Bubble Chart")
```

2. CHOROPLETH MAP

```
map("world",  
    "India",  
    fill = TRUE,  
    col = "lightgreen",  
    border = "black",  
    main = "Choropleth Map")
```

6. Network Visulation

Install packages (Run only once)

```
install.packages("igraph")
```

```
install.packages("networkD3")
```

```
install.packages("ggplot2")
```

Load libraries

```
library(igraph)
```

```
library(networkD3)
```

```
library(ggplot2)
```

1. NETWORK DIAGRAM

```
edges <- data.frame(
```

```
  from = c("HR", "HR", "IT", "Sales"),
```

```

    to = c("IT", "Sales", "Finance", "Marketing")
  )
g <- graph_from_data_frame(edges)
plot(g,
     vertex.color="skyblue",
     vertex.size=30,
     main="Network Diagram")

```

2. SANKEY DIAGRAM

```

install.packages("ggalluvial")
library(ggplot2)
library(ggalluvial)
df <- data.frame(
  From = c("HR", "HR", "IT", "Sales"),
  To = c("IT", "Sales", "Finance", "Marketing"),
  Freq = c(10, 15, 20, 25)
)
ggplot(df,
     aes(axis1=From, axis2=To, y=Freq)) +
  geom_alluvium(aes(fill=From)) +
  geom_stratum() +
  geom_text(stat="stratum", aes(label=after_stat(stratum))) +
  scale_x_discrete(limits=c("From", "To")) +
  ggtitle("Sankey Diagram")

```

3. AREA DIAGRAM

```

emp <- data.frame(
  Month = c("Jan", "Feb", "Mar", "Apr", "May"),
  Salary = c(30000, 35000, 40000, 45000, 50000)
)
ggplot(emp, aes(x=Month, y=Salary, group=1)) +
  geom_area(fill="skyblue") +

```

```
ggtitle("Area Diagram")
```

